

Grey heavy-flavour pair processes

GRACE implementation report

LaTeX edition: 30 September 2026

This document reproduces the explanation in `doc/developer_guide/pair_processes.md`, with typeset equations and a clarification of the effective-opacity convention. The build and test record below is the existing record from 29 September 2026, not a new run.

1 Scope and status

This implementation adds selectable `equilibrium` and `evolved` treatments; `legacy` remains the default. It treats *neutrino pairs*, not creation of charged muons:

- electron–positron annihilation, $e^- + e^+ \leftrightarrow \nu + \bar{\nu}$;
- nucleon bremsstrahlung, $N + N \leftrightarrow N + N + \nu + \bar{\nu}$;
- an explicitly optional transverse-plasmon model, $\gamma^* \leftrightarrow \nu + \bar{\nu}$.

These thermal channels do not require charged muons to be present. They must not be multiplied by the charged-muon abundance/density/temperature gate. Charged-current muon reactions remain the responsibility of the ordinary emission/absorption/scattering (EAS) provider. This change does not implement muon decay, muon–antimuon annihilation, flavour conversion, or missing electron-flavour pair channels in the baseline.

This is an *isotropic-kernel grey closure*, not an exact Boltzmann solution. Conservation and detailed balance can be checked independently of the accuracy of its spectral/angular assumptions. In particular, it is not a validated polar-funnel annihilation/heating model and is not claimed to outperform a spectral Monte Carlo calculation. Production use requires quadrature, resolution, timestep and target-GPU validation.

2 Configuration

Build with 3 or 5 neutrino species and initialize the repository’s existing `extern/bns_nurates` submodule. Its pinned revision is `efbabe0d4dfa6935c9ecbbd59a30c69b7cd1b91e`; no new revision is selected here.

```
m1:
  eas:
    kind: neutrino_weakhub          # alternatively neutrino_analytic
    pair_treatment: evolved        # legacy | equilibrium | evolved
    pair_quadrature_order: 16      # 8..32; compare 16, 24, 32
    weakhub_pair_content: none     # ONLY after checking table provenance
    betaeq_policy: off
    pair_annihilation: true
    bremsstrahlung: true
    plasmon_decay: true
    pair_plasmon_model: transverse_mass
```

No spectral WeakHub tables are needed: the new pair kernels are evaluated locally. The ordinary grey WeakHub table *must exclude these heavy-flavour thermal pair channels*. `weakhub_pair_content:none` is a declaration, not a filter: the code cannot subtract unknown pair contributions from an inclusive grey table. `unknown` and `included` are rejected in the new modes. The old standalone `bns_nurates` EAS adapter is not used by these modes.

The new modes reject algebraic `betaeq_policy` replacements of the actual matter state. Ordinary temperature corrections remain confined to ordinary rates. Pair rates receive neither an extra temperature-square multiplier nor an optical-depth fugacity suppression. The historical muonic $\eta = \pm 5$ clamp is retained in legacy mode only: using it in ordinary charged-current Kirchhoff factors but not in the new pair calculation would introduce incompatible equilibrium targets. GRACE's existing equation-of-state (EOS) chemical-potential convention is retained:

$$\mu_{\nu_\mu} = \mu_\mu + \mu_p - \mu_n - Q_{np}, \quad \mu_{\bar{\nu}_\mu} = -\mu_{\nu_\mu}.$$

3 Collision operator and grey equations

In the matter frame, per physical helicity species, define

$$d\Pi = D(\epsilon) d\epsilon, \quad D(\epsilon) = \frac{4\pi\epsilon^2}{(hc)^3}, \quad n = \int f d\Pi, \quad J = \int \epsilon f d\Pi.$$

Here T is expressed in energy units. $R_a(\epsilon, \epsilon')$ is the angle-independent annihilation kernel with the first argument belonging to the neutrino, the second to the antineutrino. Kernel units are $\text{cm}^3 \text{s}^{-1}$. For these thermal channels the matter bath has zero net chemical potential for the event, so

$$R_p(\epsilon, \epsilon') = e^{-(\epsilon+\epsilon')/T} R_a(\epsilon, \epsilon'). \quad (1)$$

For a specified partner distribution,

$$j_\nu(\epsilon) = \int R_p(1 - \bar{f}') d\Pi', \quad a_\nu(\epsilon) = \int R_a \bar{f}' d\Pi', \quad (2)$$

$$C_\nu = (1 - f)j_\nu - f a_\nu = j_\nu - \lambda_\nu f, \quad \lambda_\nu = j_\nu + a_\nu. \quad (3)$$

The effective opacity in the rearranged source $j - \lambda f$ is λ/c , not a/c . This does not introduce an additional physical absorption process: a is still the true inverse-reaction coefficient, and $(1 - f)j - af$ is an equally correct form. The extra j in λ accounts for own-species Pauli blocking. Both j and a here have units s^{-1} , not the units of a grey energy emissivity. The antineutrino equation uses the same event kernel with the roles of the arguments transposed. Electron-pair kernels are not generally symmetric under energy exchange.

For weights $w = 1$ (number) or $w = \epsilon$ (energy), coefficients for a specified spectral shape can be written

$$Q_w = \int w j d\Pi, \quad c\kappa_w = \frac{\int w \lambda f d\Pi}{\int w f d\Pi}. \quad (4)$$

Here Q is spontaneous emission *with partner blocking*, and own-species blocking is contained in λ . Thus both emissivity and opacity are computed from the kernel. Calling both of them independent physical inputs would also be misleading: they obey microscopic detailed balance, Eq. (1).

3.1 Equilibrium partner

Use $f^{\text{eq}} = [\exp((\epsilon - \mu)/T) + 1]^{-1}$ and the partner with $-\mu$. Then

$$j(\epsilon) = \lambda(\epsilon) f^{\text{eq}}(\epsilon), \quad B_w = \int w f^{\text{eq}} d\Pi, \quad Q_w = c\kappa_w B_w, \quad (5)$$

where the grey opacity in Eq. (4) is weighted with f^{eq} . The code evaluates both sides from the same quadrature. Equation (5) is valid here because the blocking and averaging conventions agree. It does not justify dividing an arbitrary unblocked emissivity by a degenerate Fermi–Dirac (FD) blackbody.

The independent grey sources are $S_N = Q_N - c\kappa_N n$ and $S_E = Q_E - c\kappa_E J$. They vanish at the prescribed local thermodynamic equilibrium (LTE) target but *need not give equal neutrino and antineutrino number changes away from LTE*. This is an intentional limitation of the equilibrium-partner comparison mode.

3.2 Evolved partner

At each trial of the implicit solve, GRACE’s M1 closure supplies J , H^α and Γ . The comoving number density is $n = N_{\text{lab}}/\Gamma$, with $\Gamma = W(E - v^i F_i)/J$ in the existing grey number-current closure. Reconstruct:

$$f_i = [\exp(a + b\epsilon_i/\langle\epsilon\rangle) + 1]^{-1}, \quad b > 0, \quad (6)$$

$$\sum_i W_i f_i = n, \quad \sum_i W_i \epsilon_i f_i = J.$$

The fit parameters a, b in Eq. (6) are unrelated to the absorption coefficient a_ν in Eq. (2). Both moments are matched on the *same* positive quadrature that evaluates the collisions, separately for each partner. W_i includes $D(\epsilon_i)$. This is a two-parameter FD/maximum-entropy spectral closure, not a claim that the evolved spectrum is actually thermal. Invalid or unrepresentable moments are rejected; the solver does not secretly replace the partner by matter LTE.

For every ordered pair of energy nodes, accumulate one event rate:

$$I_{ij} = W_i W_j \left\{ R_{ij}^p (1 - f_i)(1 - \bar{f}_j) - R_{ij}^a f_i \bar{f}_j \right\}, \quad (7)$$

$$S_N = \bar{S}_N = \sum_{ij} I_{ij}, \quad S_E = \sum_{ij} \epsilon_i I_{ij}, \quad \bar{S}_E = \sum_{ij} \epsilon_j I_{ij}. \quad (8)$$

Consequently the neutrino-number minus antineutrino-number source is zero for pair-only interactions. The two energies need not be equal. The whole FD family with common matter temperature and opposite chemical potentials is a pair-equilibrium family; pair interactions alone do not select beta equilibrium or erase a preexisting lepton-number difference. There is no matter-Kirchhoff reset of Eqs. (7)–(8).

The grey flux spectrum is assumed proportional to the energy spectrum. With $\lambda_E = \int \epsilon \lambda f d\Pi/J$, the covariant pair source is

$$G^\alpha = S_E u^\alpha - \lambda_E H^\alpha. \quad (9)$$

In GRACE’s undensitized Eulerian variables this contributes

$$\begin{aligned} \dot{E} &= \alpha W \left\{ S_E - \lambda_E (E - v^i F_i - J) \right\}, \\ \dot{F}_i &= \alpha (W v_i S_E - \lambda_E H_i), \quad \dot{N} = \alpha S_N. \end{aligned} \quad (10)$$

Ordinary charged-current/emission/scattering sources are added to Eq. (10) inside one coupled ten-variable diagonal implicit–explicit (IMEX) solve:

$$\begin{aligned} U - U_* - h [S_{\text{ordinary}}(U) + S_{\text{pair}}(U)] &= 0, \\ U &= (E, F_x, F_y, F_z, N, \bar{E}, \bar{F}_x, \bar{F}_y, \bar{F}_z, \bar{N}), \quad h = dt \, dt_{\text{fact}}. \end{aligned} \quad (11)$$

Matter and ordinary coefficients are held fixed over this local stage, as in the existing GRACE source treatment. This is not a simultaneous EOS+matter Newton solve. The existing conservative backreaction deposits the accepted radiation energy/momentum and lepton-number changes into the fluid. Existing backreaction limiters may throttle/reject those changes.

The solver uses scaled finite-difference Newton steps, positivity/cone checks and line search. A continuation retry solves smaller coefficients of the same original residual and ends at the full h ; it is not a sequence of hidden time substeps. Failure aborts explicitly, instead of accepting an unrelated linear fallback. A final projection makes the two solved number equations share a common pair increment; ordinary number absorption remains implicit. The full nonlinear residual is checked again after this projection. `pair_residual` records the normalized residual of the accepted state.

3.3 Aggregate species and units

Five species: indices 2/3 are separate $\nu_\mu/\bar{\nu}_\mu$; index 4 represents $\nu_\tau + \bar{\nu}_\tau$ with multiplicity 2. Three species: index 2 has multiplicity 4. Divide aggregate E, F, N and ordinary emissivities by that multiplicity before reconstructing a one-helicity spectrum. For a charge-symmetric aggregate, use $[R(\epsilon, \epsilon') + R(\epsilon', \epsilon)]/2$, evolve one representative spectrum and multiply the output by the multiplicity. Do *not* multiply its opacity by the multiplicity.

Physical sources are converted using the same constants as existing EAS:

$$\begin{aligned} J_{\text{code}} &= C_E J, & n_{\text{code}} &= C_N n, \\ Q_{E,\text{code}} &= C_E Q_E / \text{TIMEGF}, & Q_{N,\text{code}} &= C_N Q_N / \text{TIMEGF}, \\ \lambda_{\text{code}} &= \lambda / \text{TIMEGF}. \end{aligned}$$

Here $C_E = \text{mev_to_erg} \text{RHOGF EPSGF}$ and $C_N = \text{mnuc_cgs} \text{RHOGF}$. Conserved variables are divided by $\sqrt{\gamma}$ before solving, multiplied back afterwards.

4 Kernel implementation and approximations

Electron pairs use the pinned BNS_NURATES Pons/Bruenn PairPsi kernel, including its relativistic/massless-electron approximation (not an exact finite-electron-mass kernel at arbitrarily low temperature), algebraically evaluating its absorption coefficient without the unstable product $\exp[(\epsilon + \epsilon')/T] R_p$. In the library's symbols the implemented expression is

$$\begin{aligned} R_a^{ee} &= -\frac{CT^2}{2[1 - e^{-(y+z)}]} \left[\alpha_1^2 \Psi_0(y, z, \eta_e) + \alpha_2^2 \Psi_0(z, y, \eta_e) \right], \\ y &= \epsilon/T, \quad z = \epsilon'/T, \end{aligned}$$

with $C = \text{kBS_Pair_Phi}$ and the heavy-flavour weak couplings from the pinned library; the result is converted from nm^3/s to cm^3/s . NN pairs use its unchanged Hannestad–Raffelt 1998 kernel, without the optional Fischer medium correction. BNS_NURATES uses nm units: the numerical baryon density in nm^{-3} is its value in cm^{-3} multiplied by 10^{-21} , and the numerical kernel in cm^3/s is its value in nm^3/s multiplied by 10^{-21} . The electron-pair expression retains the library's numerical nonnegative-rate guard; this is not an independent revalidation of the library's fit over every EOS state. Extremely degenerate states require explicit convergence checks.

The optional plasma approximation uses two transverse polarisations, residue $Z = 1$, $\omega^2 = k^2 + m_\gamma^2$,

$$\begin{aligned} m_\gamma^2 &= \frac{4\alpha_{\text{EM}}}{3\pi} \left(\mu_e^2 + \pi^2 T^2/3 \right), \\ \Gamma_T(\omega) &= \frac{G_F^2 C_V^2 m_\gamma^6}{48\pi^2 \alpha_{\text{EM}} \hbar \omega}, \quad C_V = -\frac{1}{2} + 2 \sin^2 \theta_W. \end{aligned} \tag{12}$$

For $\omega = \epsilon + \epsilon'$ and $k^2 = \omega^2 - m_\gamma^2$, the allowed region is $(\epsilon - \epsilon')^2 \leq k^2$, with the transverse decay distribution $dP/d\epsilon = 3[1 + (\epsilon - \epsilon')^2/k^2]/(4k)$. Transforming the thermal photon phase space gives the joint event kernel

$$D(\epsilon)D(\epsilon')R_a = 2 \frac{4\pi}{(hc)^3} \omega \Gamma_T(\omega) [1 + f_{\text{BE}}(\omega)] \frac{3}{4} \left[1 + \frac{(\epsilon - \epsilon')^2}{k^2} \right]. \tag{13}$$

$R_p = \exp(-\omega/T)R_a$ includes f_{BE} ; the inverse rate includes $1 + f_{\text{BE}}$, not $1 - f_{\text{BE}}$. This approximation neglects longitudinal/axial/mixed modes, charged-muon polarisation, finite electron mass corrections and the momentum-dependent transverse dispersion. It is not quantitatively reliable for all cool/degenerate states; opt-in is mandatory. Plasma thresholds can converge more slowly than smooth ee/NN kernels on this tensor quadrature.

5 EAS and evolution workflows

5.1 Equilibrium mode

1. Read EOS and ordinary analytic/WeakHub rates; suppress old thermal extras.
2. Use the consistent muonic chemical potentials in ordinary Kirchhoff factors; apply the existing *ordinary* gate and temperature corrections.
3. Construct thermal pair kernels and opposite-chemical-potential LTE spectra.
4. Compute both grey Q and stimulated-absorption κ from Eqs. (2)–(5).
5. Add these pair terms after ordinary corrections; preserve scattering.
6. Use the existing per-species E, F, N implicit update and fluid backreaction.

5.2 Evolved mode

1. Perform the same ordinary EAS calculation, excluding old thermal extras.
2. Store physical T, μ_e, n_b, X_n, X_p and enabled channels in pair auxiliaries. Ordinary EAS slots deliberately contain *no pair terms*.
3. In each implicit stage construct/cache one kernel matrix per physical pair.
4. Reconstruct both spectra from each trial $J, N/\Gamma$; compute Eqs. (7)–(10).
5. Solve Eq. (11), including ordinary EAS, simultaneously for the two partners. Treat the charge-symmetric residual heavy aggregate with its multiplicity.
6. Run unchanged electron/photon source updates, then existing backreaction.

Equilibrium-blended initial data temporarily use equilibrium pair EAS even when the simulation selects evolved partners. The usual final EAS pass then restores ordinary-only slots and material snapshots for evolution. Atmosphere cells disable pair channels and clear their material/residual auxiliaries.

6 Code map, review and validation

- `neutrino_pair_collision.hh`: quadrature, kernels, FD reconstruction, common blocked collision integrals and units.
- `neutrino_pair_update.hh`: coupled M1 residual and local nonlinear solver.
- `eas_kinds.hh`, `parameters/m1.yaml`: setup selectors and incompatible-mode checks.
- `eas_policies.hh`: ordinary/pair separation, matching muonic LTE targets, equilibrium coefficients and evolved material snapshots.
- `variable_indices.hh/.cpp`: material/channel/residual auxiliary registration.

- `m1.hh`: one-cell species mapping, densitization and aggregate factors.
- `evolve.cpp`: pair-stage launch before ordinary sources/backreaction.
- `m1.cpp`: setup validation, small-tile EAS launch and initialization support.
- `CMakeLists.txt`, `test/CMakeLists.txt`, `test_m1_pairs.cpp`: headers and tests. The Kokkos test helper now propagates the bundled YAML/p4est/HDF5 dependencies; existing backreaction tests support both Cowling and Z4 metric layouts. Two legacy-rate assertions were corrected to test the already-existing smooth muon suppression factor, rather than assuming a hard cutoff. The legacy rate implementation itself was not changed for those assertions.

The hot path has fixed-size automatic storage and Kokkos device functions, no host-only solver, dynamic allocation or CUDA/HIP-specific API. Kernel matrices are computed once per stage, not once per Newton trial. Separate small tiles avoid the ordinary kernel’s GPU launch-bound assumptions. This is portable source design, not a GPU performance measurement; register pressure, spills and throughput still need measurement on each target backend.

Tests cover unit normalization, FD moments, kernel orientation, LTE balance, blocking/vacuum emission, unequal populations, implicit stiffness, simultaneous ordinary EAS, moving-fluid LTE and execution on the configured Kokkos backend. Run

```
ctest --test-dir build-pairs-cpu -R m1_pair_test --output-on-failure
```

Do not equate unit-test success with a validated neutron-star evolution.

6.1 Verified build (2026-09-29)

The complete `grace` executable built with GCC/WSL, Kokkos Serial, Debug, 3 spatial dimensions, 5 neutrino species, muons enabled and Cowling metric. The following suites passed after the review/fix cycles:

Suite	Cases	Assertions
<code>m1_pair_test</code>	9	119
<code>m1_analytic_rates_test</code>	33	1655
<code>m1_backreaction_test</code>	44	643
Total	86	2417

```
cmake -S . -B build-pairs-cpu -DGRACE_USE_BUNDLED_DEPS=ON \
  -DGRACE_M1_NU_SPECIES=5 -DGRACE_3D=ON -DGRACE_METRIC_EVOL=COWLING \
  -DGRACE_PERF_TUNING=generic -DCMAKE_BUILD_TYPE=Debug
cmake --build build-pairs-cpu --target grace m1_pair_test \
  m1_analytic_rates_test m1_backreaction_test -j6
ctest --test-dir build-pairs-cpu \
  -R 'm1_(pair|analytic_rates|backreaction)_test$' --output-on-failure
```

Not verified here: GPU compilation/performance, a three-species executable, full coupled hydrodynamic/merger evolutions, all extreme EOS states, or quantitative accuracy against full angular/spectral transport. The standard M1 flux cone is checked; this is not a full fermionic angular-realizability closure. Even with matched chemical potentials, differences between finite pair quadrature and the existing ordinary-EAS Fermi-integral fits leave a quadrature-level LTE mismatch: check order convergence before production. The existing generic implicit-solver counters describe the ordinary solver; `pair_residual` is the new coupled-pair solver’s diagnostic.

7 Literature and what can actually be criticized

Ng et al., [WeakHub](#), give the pair collision operator and reaction kernels (Sec. 3.4). [Chiesa et al.](#), [BNS_NURATES](#), describe the portable kernel library and reconstruction from grey moments. The new code reuses the kernels, not the library’s complete grey closure. [Braaten and Segel](#) give the more complete plasma treatment; the simplified model (12)–(13) must not be confused with it.

[Gieg et al.](#), Appendix A, drop neutrino blocking in A4 before writing FD detailed balance in A5. That cannot be the exact identity of the fully blocked collision operator; an additional closure choice is involved. This implementation keeps both blocking factors instead. [Foucart](#), Sec. II.4, uses evolved spectral estimates with an opaque-region equilibrium fallback and an angular correction; it is not simply an equilibrium-partner treatment. A blanket claim that this new grey R_0 implementation is more accurate is unwarranted. No unpublished criticism attributed to Harry is treated as evidence of a specific error.